

WAD+CC Viva QnA

1. HTML / CSS / Bootstrap Viva

Basic HTML Questions

Q1. What is HTML?

HTML stands for HyperText Markup Language.

It is used to create the structure of web pages.

Q2. What are HTML tags?

Tags are predefined keywords enclosed in angle brackets used to define webpage elements.

Example:

```
<h1>Hello</h1>
```

```
<p>Paragraph</p>
```

Q3. Difference between HTML and HTML5?

HTML	HTML5
Older version	Latest version
No semantic tags	Semantic tags available
No audio/video support	Supports audio/video
Uses plugins	Native multimedia support

Q4. What are semantic tags?

Semantic tags clearly describe their meaning.

Examples:

```
<header>
```

```
<nav>
```

```
<section>
```

```
<footer>
```

```
<article>
```

Q5. Difference between div and span?

div	span
Block element	Inline element
Takes full width	Takes required width

Q6. What is the use of form tag?

Used to collect user input.

Example:

```
<form>
```

```
<input type="text">
```

```
</form>
```

Q7. What are input types?

- text
- password
- email
- radio
- checkbox
- file
- submit

Q8. Difference between id and class?

id	class
-----------	--------------

Unique	Multiple elements
--------	-------------------

# selector	. selector
------------	------------

CSS Viva**Q9. What is CSS?**

CSS stands for Cascading Style Sheets.

Used to style HTML pages.

Q10. Types of CSS?

1. Inline CSS
2. Internal CSS
3. External CSS

Q11. Difference between margin and padding?

Margin	Padding
---------------	----------------

Outside border	Inside border
----------------	---------------

Q12. What is Flexbox?

CSS layout model used for alignment and spacing.

Example:

display:flex;

justify-content:center;

align-items:center;

Q13. What is CSS Grid?

2D layout system for rows and columns.

Q14. What is media query?

Used for responsive design.

Example:

@media screen and (max-width:768px)

Q15. What is z-index?

Controls stack order of elements.

Higher z-index appears on top.

Bootstrap Viva

Q16. What is Bootstrap?

Bootstrap is a CSS framework used to create responsive websites quickly.

Official website: [Bootstrap](https://getbootstrap.com/)

Q17. What is responsive web design?

Website automatically adjusts according to screen size.

Q18. Explain Bootstrap Grid System.

Bootstrap uses 12-column grid system.

Example:

col-md-6

Means 6 columns on medium screens.

Q19. Difference between container and container-fluid?

container	container-fluid
------------------	------------------------

Fixed width	Full width
-------------	------------

Q20. What are Bootstrap cards?

Flexible content containers.

2. JavaScript / AJAX / Local Storage Viva

Q21. Difference between Java and JavaScript?

Java	JavaScript
Programming language	Scripting language
Compiled	Interpreted

Q22. What is JavaScript?

Client-side scripting language used for webpage interactivity.

Q23. What is DOM?

DOM stands for Document Object Model.

JavaScript can manipulate HTML elements using DOM.

Q24. Difference between let, var, const?

var	let	const
Function scope	Block scope	Block scope
Re-declaration allowed	Not allowed	Not allowed

Q25. What is an array?

Collection of elements.

```
let arr=[1,2,3];
```

Q26. What is localStorage?

Browser storage used to store data permanently.

Methods:

```
localStorage.setItem()
```

```
localStorage.getItem()
```

Q27. Difference between sessionStorage and localStorage?

sessionStorage	localStorage
Temporary	Permanent
Deleted after tab close	Remains stored

Q28. What is AJAX?

AJAX = Asynchronous JavaScript and XML.

Used to send requests without refreshing page.

Q29. What are AJAX methods?

- GET
- POST
- PUT
- DELETE

Q30. Difference between GET and POST?

GET	POST
Retrieves data	Sends data
Visible in URL	Hidden

3. Git & GitHub Viva

Q31. What is Git?

Git is a distributed version control system.

Official site: [Git](https://git-scm.com/)

Q32. What is GitHub?

GitHub is cloud platform for hosting Git repositories.

Official site: [GitHub](#)

Q33. Important Git Commands?

git init

git status

git add .

git commit -m "message"

git push origin main

git pull origin main

Q34. Difference between Git and GitHub?

Git	GitHub
Tool	Cloud platform

Q35. What is repository?

Storage location for project code.

Q36. What is commit?

Snapshot of project changes.

Q37. What is branch?

Separate development line in Git.

4. Docker Viva**Q38. What is Docker?**

Docker is containerization platform used to package applications.

Official site: [Docker](#)

Q39. Difference between VM and Docker?

VM	Docker
Heavy	Lightweight
Separate OS	Shares host OS

Q40. What is container?

Running instance of Docker image.

Q41. What is Docker Image?

Blueprint/template for container.

Q42. Common Docker Commands

`docker --version`

`docker pull ubuntu`

`docker images`

`docker run ubuntu`

`docker ps`

`docker stop container_id`

Q43. What is Dockerfile?

Text file containing Docker instructions.

5. Angular Viva**Q44. What is Angular?**

Angular is TypeScript-based frontend framework.

Official site: [Angular](https://angular.io/)

Q45. What is TypeScript?

Superset of JavaScript with static typing.

Q46. Angular CLI commands?

`ng new appname`

`ng serve`

`ng generate component login`

Q47. What is component in Angular?

Building block of Angular application.

Q48. What is data binding?

Connecting data between TS and HTML.

Types:

- Interpolation
- Property binding
- Event binding
- Two-way binding

Q49. What is two-way binding?

Synchronizes data between UI and TS.

`[(ngModel)]`

Q50. What is routing?

Navigation between components/pages.

Q51. What is service in Angular?

Reusable logic shared across components.

6. Node.js / Express Viva

Q52. What is Node.js?

Runtime environment used to run JavaScript on server.

Official site: [Node.js](https://nodejs.org/)

Q53. Advantages of Node.js?

- Fast
- Event-driven
- Non-blocking
- Scalable

Q54. What is npm?

Node Package Manager.

Q55. What is Express.js?

Node.js framework used to build web applications and APIs.

Official site: [Express.js](https://expressjs.com/)

Q56. What is package.json?

Project configuration file.

Q57. What is middleware?

Function executed between request and response.

Q58. What are CRUD operations?

- Create
- Read
- Update
- Delete

Q59. REST API methods?

Method	Operation
--------	-----------

GET	Read
-----	------

POST	Create
------	--------

PUT	Update
-----	--------

DELETE	Delete
--------	--------

7. MongoDB Viva

Q60. What is MongoDB?

NoSQL document database.

Official site: [MongoDB](https://www.mongodb.com/)

Q61. Difference between SQL and MongoDB?

SQL MongoDB

Tables Collections

Rows Documents

Q62. What is document in MongoDB?

Data stored in JSON-like format.

Q63. Common MongoDB Commands

show dbs

use student

db.createCollection("users")

db.users.find()

8. jQuery / jQuery Mobile Viva

Q64. What is jQuery?

JavaScript library simplifying DOM operations.

Official site: [jQuery](https://jquery.com/)

Q65. What is jQuery Mobile?

Framework for mobile-friendly web apps.

Official site: [jQuery Mobile](https://jquerymobile.com/)

Q66. What is selector in jQuery?

Used to select HTML elements.

Example:

`$("#id")`

`$(".class")`

Q67. Difference between JavaScript and jQuery?

jQuery simplifies JavaScript coding.

9. AWS / EC2 / Apache Viva

Q68. What is Cloud Computing?

Delivery of computing services over internet.

Q69. Types of cloud?

- Public

- Private
- Hybrid

Q70. What is AWS?

Amazon Web Services cloud platform.

Official site: [Amazon Web Services \(AWS\)](https://aws.amazon.com/)

Q71. What is EC2?

Elastic Compute Cloud — virtual server service in AWS.

Amazon EC2

Q72. What is Ubuntu?

Linux-based operating system.

Q73. What is Apache Web Server?

Software used to host websites.

Official site: [Apache HTTP Server](https://httpd.apache.org/)

Q74. Apache install commands?

sudo apt update

sudo apt install apache2

sudo systemctl start apache2

sudo systemctl enable apache2

Q75. What is public IP?

IP accessible over internet.

Q76. What is Security Group in AWS?

Acts like firewall controlling inbound/outbound traffic.

Q77. Which ports are opened for website hosting?

- 22 → SSH
- 80 → HTTP
- 443 → HTTPS

Q78. Steps to connect EC2?

ssh -i key.pem ubuntu@public-ip

10. Virtual Machine / Vagrant Viva

Q79. What is Virtual Machine?

Software-based computer system.

Q80. What is hypervisor?

Software managing virtual machines.

Examples:

- VirtualBox
- VMware

Q81. What is Vagrant?

Tool for automating VM setup.

Official site: [Vagrant](https://www.vagrantup.com/)

Q82. Vagrant Commands?

vagrant init

vagrant up

vagrant ssh

vagrant halt

Q83. Difference between VM and Container?

VM	Container
Full OS	Shared OS

11. Google App Engine Viva**Q84. What is Google App Engine?**

Platform as a Service (PaaS) by Google Cloud.

Official site: [Google App Engine](https://cloud.google.com/appengine/)

Q85. Difference between IaaS, PaaS, SaaS?**Type Meaning**

IaaS Infrastructure

PaaS Platform

SaaS Software

12. Salesforce Viva**Q86. What is Salesforce?**

Cloud-based CRM platform.

Official site: [Salesforce](https://www.salesforce.com/)

Q87. What is CRM?

Customer Relationship Management.

Q88. What is object in Salesforce?

Database table equivalent.

Q89. What is field in Salesforce?

Column equivalent.

Q90. Difference between standard and custom object?

Standard	Custom
-----------------	---------------

Predefined	User-created
------------	--------------

13. Networking & Ports Viva

Q91. What is IP address?

Unique address identifying device on network.

Q92. What is localhost?

Refers to current machine.

IP:

127.0.0.1

Q94. What is HTTP?

Protocol used for web communication.

Q95. Difference between HTTP and HTTPS?

HTTP	HTTPS
-------------	--------------

Not secure	Secure
------------	--------

Port 80	Port 443
---------	----------

Q96. What is DNS?

Domain Name System converts domain to IP.

14. Important Difficult Viva Questions

Q97. Why Node.js is called non-blocking?

Because it uses asynchronous event-driven architecture.

Q98. Why MongoDB is called NoSQL?

Because it stores unstructured document-based data instead of relational tables.

Q99. Why Bootstrap is preferred?

- Faster development
- Responsive
- Predefined components

Q100. Explain MVC architecture.

- Model → Data
- View → UI
- Controller → Logic

Q101. Difference between authentication and authorization?

Authentication Authorization

Identity verification Access control

Q102. Difference between frontend and backend?

Frontend Backend

User interface Server-side logic

Q103. Explain API.

Application Programming Interface allowing communication between applications.

Q104. What happens when you type URL in browser?

1. DNS lookup
2. Server request
3. Response sent
4. Browser renders page

VERY IMPORTANT PRACTICAL-BASED QUESTIONS

Q105. Why did you use Express in CRUD API?

Because Express simplifies route handling and server creation.

Q106. Why MongoDB for CRUD practical?

Because MongoDB stores JSON-like documents which integrate well with Node.js.

Q107. Why use Bootstrap in dashboard practical?

To make webpage responsive quickly.

Q108. Why use localStorage in AJAX practical?

To store user data permanently in browser.

Q109. Why use EC2 instance?

To host application on cloud virtual server.

Q110. Why use Docker?

To ensure application runs consistently in every environment.

Linux Commands Viva Table

Command	Full Form / Syntax	Purpose
pwd	Present Working Directory	Shows current directory path
ls	List	Lists files and folders
cd	Change Directory	Changes folder/directory
mkdir foldername	Make Directory	Creates new folder
touch filename	—	Creates empty file
rm filename	Remove	Deletes file
cp source destination	Copy	Copies file/folder
mv old new	Move	Moves or renames file
sudo command	Super User Do	Runs command as administrator
clear	—	Clears terminal screen
history	—	Shows previously executed commands
man command	Manual	Displays help/manual of command
ifconfig	Interface Configuration	Shows IP/network details
ping website	Packet Internet Groper	Checks internet/network connectivity

Package / Software Commands

Command	Purpose
sudo apt update	Updates package list
sudo apt install apache2	Installs Apache server
node -v	Shows Node.js version
npm -v	Shows npm version
git --version	Shows Git version
python --version	Shows Python version

Git Commands Viva Table

Command	Purpose
git init	Initializes Git repository
git status	Shows repository status
git add .	Adds files to staging area

Command	Purpose
git commit -m "msg"	Saves changes with message
git push origin main	Uploads code to GitHub
git pull origin main	Downloads latest code
git clone url	Copies repository from GitHub

Node.js / npm Commands

Command	Purpose
npm init	Creates package.json
npm install express	Installs Express package
node index.js	Runs Node.js application
npm start	Starts application
npm install	Installs all dependencies

Angular Commands

Command	Purpose
ng new appname	Creates Angular project
ng serve	Runs Angular server
ng generate component login	Creates new component
ng build	Builds Angular project

Docker Commands

Command	Purpose
docker --version	Shows Docker version
docker pull ubuntu	Downloads Docker image
docker images	Lists Docker images
docker run ubuntu	Runs container
docker ps	Shows running containers

Command	Purpose
<code>docker stop id</code>	Stops container
<code>docker build .</code>	Builds Docker image

MongoDB Commands

Command	Purpose
<code>show dbs</code>	Shows all databases
<code>use mydb</code>	Selects database
<code>db.createCollection()</code>	Creates collection
<code>db.users.find()</code>	Displays documents
<code>db.users.insertOne()</code>	Inserts document
<code>db.users.updateOne()</code>	Updates document
<code>db.users.deleteOne()</code>	Deletes document

Apache / AWS Commands

Command	Purpose
<code>sudo apt install apache2</code>	Installs Apache
<code>sudo systemctl start apache2</code>	Starts Apache server
<code>sudo systemctl enable apache2</code>	Starts Apache at boot
<code>sudo systemctl status apache2</code>	Shows Apache status
<code>ssh -i key.pem ubuntu@ip</code>	Connects EC2 instance

Vagrant Commands

Command	Purpose
<code>vagrant init</code>	Initializes Vagrant
<code>vagrant up</code>	Starts virtual machine
<code>vagrant ssh</code>	Connects VM

Command	Purpose
vagrant halt	Stops VM

HTTP Methods Viva Table

Method	Purpose
GET	Retrieve data
POST	Send/Create data
PUT	Update data
DELETE	Delete data

Important Port Numbers

Port	Service
20/21	FTP
22	SSH
23	Telnet
25	SMTP
53	DNS
67/68	DHCP
80	HTTP
110	POP3
143	IMAP
443	HTTPS
3306	MySQL
27017	MongoDB

Why key pair is used in EC2?

For secure SSH login authentication.

Why security groups are important?

They act as firewall.

What is inbound rule?

Incoming traffic rules.

Why open port 22?

For SSH connection.

Why open port 80?

For website hosting.

Virtualization Concepts

Questions:

What is hypervisor?

Software managing virtual machines.

Types:

- Type 1
- Type 2

Examples:

- VMware
- VirtualBox